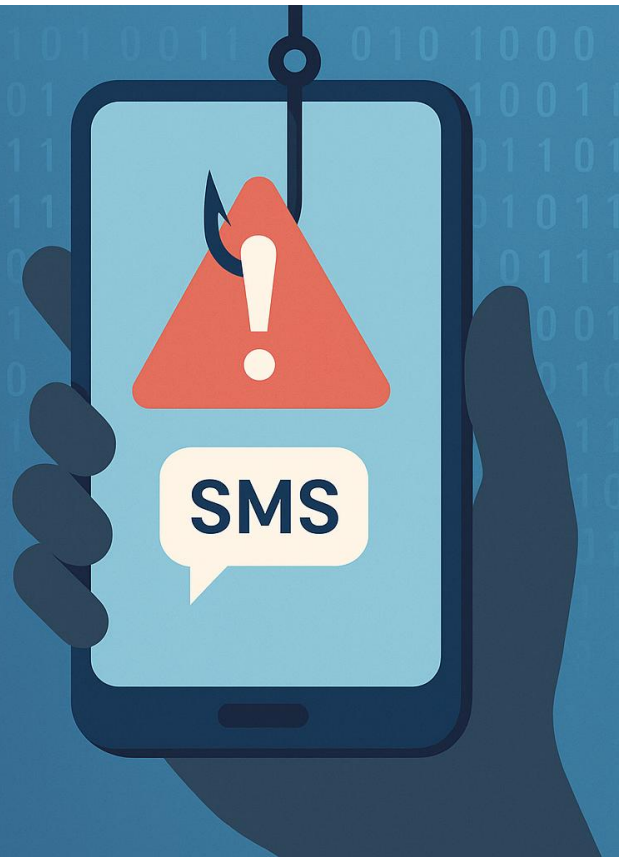


DECODING DECEPTION

IDENTIFYING SMS-BASED CYBER ATTACKS



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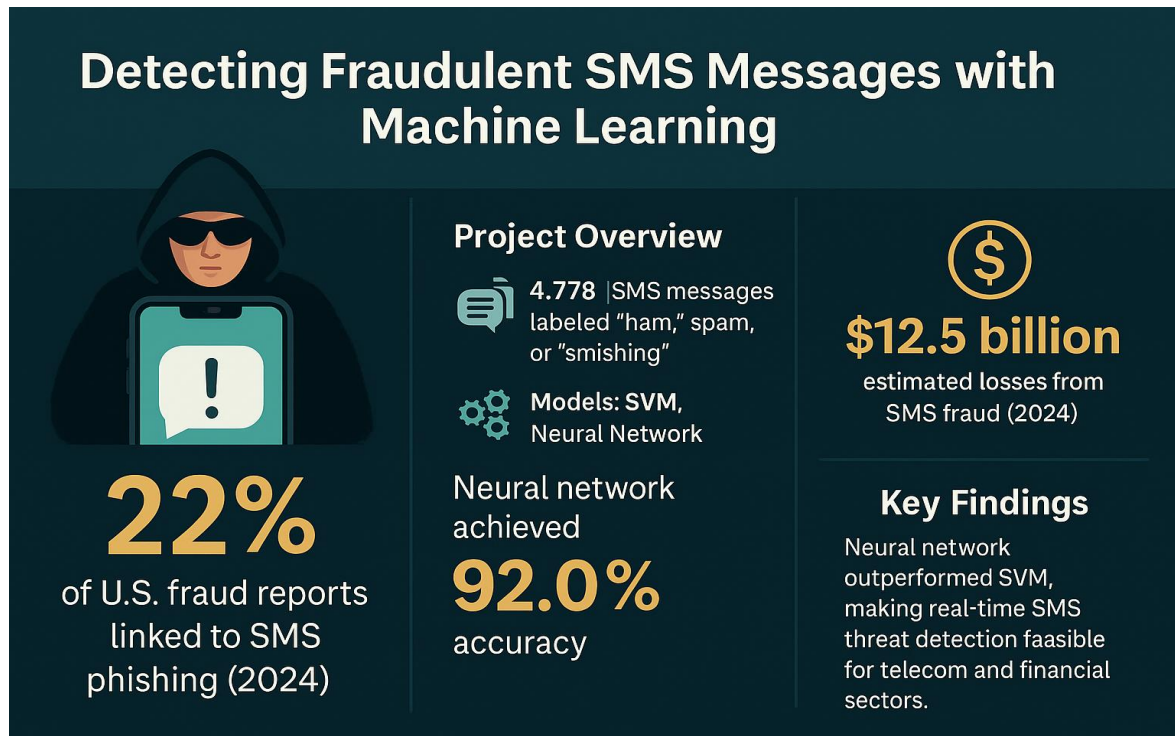
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1. Executive Summary



The rise of mobile communication has introduced new vulnerabilities in the digital ecosystem, particularly through SMS-based phishing attacks, commonly known as *smishing*. These deceptive messages often mimic trusted entities and exploit urgency to manipulate recipients into divulging sensitive personal information.

In response to this growing threat responsible for over 22% of U.S. fraud reports and \$12.5 billion in estimated losses in 2024 our project presents a machine learning-based solution for detecting and classifying fraudulent SMS messages. We analyzed a dataset of 4,778 messages, labeled as *ham*, *spam*, or *smishing*, using advanced text vectorization (TF-IDF) and engineered metadata features (presence of EMAIL, PHONE, and URL).

Two classification models were implemented and evaluated:

- A Support Vector Machine (SVM) with a linear kernel
- A feedforward Neural Network with optimized hidden layers

While both models achieved high performance, the neural network consistently outperformed the SVM particularly in detecting smishing achieving an accuracy of 92.03% and a Kappa statistic of 0.756, indicating strong interclass reliability.

These results support the deployment of deep learning models in real-world settings such as telecommunications, banking, and mobile security, where real-time detection of malicious SMS traffic is crucial to prevent fraud, identity theft, and financial exploitation.

2. Data Understanding and Preparation

The dataset used in this project consists of 4,778 SMS messages, each labeled into one of three categories:

- Ham: Legitimate and benign text messages.
- Spam: Unsolicited messages, typically promotional.
- Smishing: SMS-based phishing attempts aimed at deceiving users into revealing sensitive personal information.

This structured labeling provides a clear foundation for supervised learning, enabling models to distinguish between normal user behavior and deceptive messaging patterns. By defining these three distinct classes, the dataset captures a broad spectrum of communication styles from benign conversational messages to malicious attempts at exploitation, making it ideal for training and validating machine learning models focused on real-world SMS fraud detection.

Variable Overview

All variables in the dataset are categorical (qualitative) in nature:

Variable	Type	Description
LABEL	Target	Categorical response variable with 3 levels: ham, spam, and smishing.
EMAIL	Binary	Indicates if the message contains an email address (Yes/No).
PHONE	Binary	Indicates if the message contains a phone number (Yes/No).

URL	Binary	Indicates if the message includes a URL (Yes/No).
TEXT	Textual	The raw body of the SMS message.

The design of the dataset centered around categorical predictors such as EMAIL, PHONE, and URL strategically captures structural traits commonly associated with fraudulent messages. While these binary flags may seem simplistic, their presence or absence provides critical signals about the intent and anatomy of a message. For instance, smishing and spam texts often include embedded phone numbers or URLs to lure users into responding or clicking malicious links, making these structural cues valuable for early detection.

When these structural indicators are combined with the semantic richness extracted from the TEXT field, a layered representation of each message emerges. This dual-pronged approach leveraging both surface-level metadata and deeper linguistic patterns empowers machine learning models to recognize fraud even in adversarial scenarios where scammers attempt to disguise their tactics. By treating every message as a composite of structure and content, the dataset supports a nuanced understanding of digital deception and lays the groundwork for robust, real-time SMS fraud detection systems.

Data Cleaning and Preprocessing

Robust preprocessing is essential to ensure the quality and consistency of inputs used in machine learning workflows. The following steps were performed:

- **Missing Value Treatment:** The dataset was cleaned using `na.omit()` in R, ensuring that all entries used for modeling were complete.
- **Label Normalization:** The LABEL column was standardized to lowercase to remove inconsistencies due to case variations.
- **Text Normalization:**
 - Conversion to lowercase
 - Removal of punctuation and numeric characters
 - Stopword filtering using standard English stopwords dictionaries
- **Tokenization and Feature Transformation:**

- SMS text was converted into structured features using TF-IDF (Term Frequency–Inverse Document Frequency).
- This transformation captures importance by penalizing frequent but non-informative words and amplifying domain-specific scam cues.

Please Note: A high-sparsity Document-Term Matrix (DTM) was generated and pruned using a sparsity threshold of 99% to retain informative tokens while reducing computational noise.

Effective machine learning begins long before model selection starts with high-quality data preprocessing. In this project, we approached data cleansing not merely as a technical requirement, but as a crucial step in preserving the integrity and signal strength of the dataset. The application of `na.omit()` in R ensured that our modeling pipeline was built exclusively on complete records, eliminating the risk of training on partial or inconsistent examples. Likewise, standardizing the LABEL variable to lowercase eliminated superficial distinctions that could fragment class labels and distort learning.

The raw text messages, while rich in meaning, needed substantial refinement to be rendered machine-readable. Through a series of linguistic normalization steps—lowercasing, punctuation and digit removal, and stop-word filtering—we stripped the text down to its essential elements. These transformations help models avoid being distracted by irrelevant patterns such as capitalization or filler words and instead focus on the meaningful core of a message. The process not only improves learning efficiency but also helps in generalizing better to unseen data, particularly important when working with noisy, user-generated text.

To capture the true voice of fraudulent communication, we employed TF-IDF vectorization, a technique that assigns weight to words based on their frequency in a message relative to their overall frequency across the corpus. This method proved especially valuable in highlighting scam-related keywords like “verify,” “account,” or “urgent,” which occur disproportionately in smishing texts. The resulting Document-Term Matrix (DTM) was then pruned using a sparsity threshold of 99%, retaining only

the most informative linguistic signals. This ensured that our models were trained not just in words, but on words that matter those that scammers strategically deploy to deceive users.

3. Exploratory Data Analysis (EDA)

Univariate Distribution

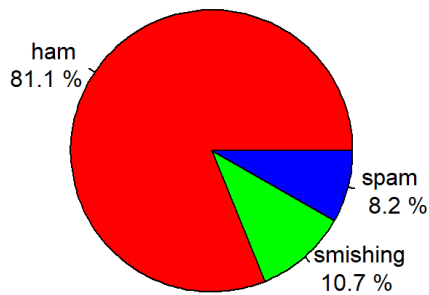
To understand the baseline structure of the dataset, frequency distributions and pie charts were generated for each of the categorical variables.

- LABEL Distribution:
 - Ham: 81.1% of the dataset
 - Smishing: 10.8%
 - Spam: 8.1%

An initial examination of the LABEL variable reveals a highly imbalanced distribution among the three message classes. Ham messages those classified as legitimate and non-malicious comprise a dominant 81.1% of the dataset. In contrast, smishing messages account for 10.8%, and spam messages make up the remaining 8.1%. This distribution aligns closely with typical user behavior in mobile communications, where most SMS exchanges are legitimate, personal, or transactional in nature.

While this imbalance is reflective of real-world proportions, it poses a critical challenge for machine learning: the minority classes, particularly smishing, are the most harmful and costly when misclassified. A naïve model could achieve high overall accuracy by simply predicting “ham” for most messages yet fail completely at detecting actual threats. As such, special care must be taken during model training through techniques like stratified sampling, cost-sensitive learning, or class-specific metrics to ensure that the classifier prioritizes recall for smishing without being overwhelmed by the abundance of benign messages.

Distribution of LABEL in BAF



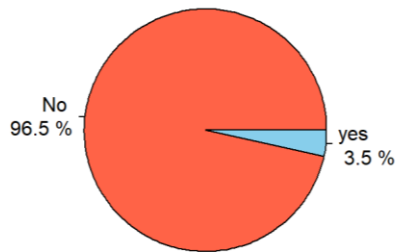
- Metadata Variables:
 - PHONE: Present in 14.1% of messages
 - URL: Present in only 3.5% of messages
 - EMAIL: Present in just 0.3% of messages, indicating limited utility

Exploring the distribution of binary metadata features reveals varying levels of signal strength across the predictors. PHONE numbers are the most common metadata cue, appearing in 14.1% of all messages. URLs are significantly less frequent, present in just 3.5% of messages. EMAIL addresses, however, are practically absent, occurring in only 0.3% of the dataset suggesting minimal relevance for downstream modeling. These metrics provide an important window into how scammers structure their messages to include deceptive content beyond just words.

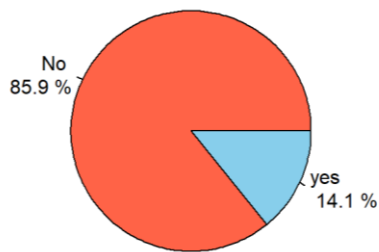
These metadata features act as structural markers that differentiate malicious messages from benign ones. The relatively higher presence of PHONE numbers in smishing and spam messages supports the hypothesis that fraudsters often prompt recipients to call or text back as part of the scam. Similarly, even though URLs are sparse, their inclusion is often tightly linked to phishing attempts, where a single malicious link can lead to credential theft or financial exploitation. Conversely, the negligible use of EMAIL addresses in SMS content indicates that this variable contributes little discriminatory power and might be excluded or down weighted in feature selection. Together, these

variables enhance a model’s capacity to identify risky message structures even in the absence of overt language patterns.

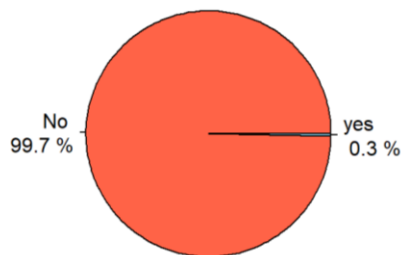
URL Presence in BAF



PHONE Presence in BAF



Email Presence in BAF



Bivariate Analysis

Given that all features are qualitative, bivariate relationships were explored by examining the relative distribution of binary predictor variables (EMAIL, PHONE, URL) across each class of the response variable (LABEL).

The resulting stacked bar plots revealed the following insights:

- Over 60% of smishing and spam messages contained either a phone number or URL.
- Ham messages rarely included these elements, affirming their discriminative power.
- The presence of email addresses was found to be uniformly rare across all classes, suggesting it holds low predictive relevance.

These bivariate relationships provide strong evidence that fraudulent SMS messages often follow distinct structural patterns. The concentration of PHONE numbers and URLs in smishing and spam messages reflects the operational mechanics of SMS-based fraud, where recipients are lured into clicking malicious links or initiating contact with attackers through deceptive call-to-actions. The sharp contrast with ham messages where such elements are notably absent reinforces their value as reliable discriminatory features.

Moreover, this structural divergence mirrors real-world cybercrime tactics. Smishing campaigns frequently mimic trusted entities like banks or delivery services, prompting users to "verify accounts" or "resolve issues" via included links or phone numbers. The use of these embedded elements not only heightens the psychological urgency of the message but also creates direct exploitation pathways, such as credential harvesting or social engineering via callback scams.

Interestingly, the scarcity of email addresses across all classes suggests an adaptation in attacker behavior, possibly influenced by user habits or spam filters that detect traditional email-based phishing more effectively. As a result, cybercriminals may be favoring

URLs and phone numbers in SMS fraud, which are harder to vet in real time and are more familiar within the mobile communication context. This evolution underscores the importance of continuously adapting detection models to align with emerging threat patterns.

4. Feature Engineering and TF-IDF Analysis

Structural Relationships Between Variables

Since all variables in this dataset are qualitative (categorical), the relationships between the response variable (LABEL) and the predictor variables (EMAIL, PHONE, URL) were examined through cross-tabulations and stacked bar visualizations. This analysis allowed us to assess how the presence of specific structural features varies across message classes.

Observations:

- Smishing and spam messages frequently include phone numbers and URLs, with over 60% containing at least one of these markers.
- In contrast, ham messages are rarely accompanied by such structural elements, making the absence of these features a strong signal of legitimacy.
- Email addresses were consistently rare across all message types, appearing in fewer than 0.3% of the dataset, thereby contributing minimal predictive value.

To evaluate how structural indicators relate to message classification, we conducted cross-tabulations and visualized the relative frequency of each binary predictor variable EMAIL, PHONE, and URL across the three LABEL categories: ham, spam, and smishing. This approach helped illuminate distinct usage patterns of metadata elements in legitimate versus malicious SMS content. The results were revealing more than 60% of spam and smishing messages contained either a phone number or a URL, whereas such features were scarcely found in ham messages.

This pronounced contrast highlights the predictive value of structural cues in fraud detection. The inclusion of phone numbers and URLs appears to be a deliberate strategy by cybercriminals to establish a false sense of legitimacy or urgency, prompting recipients to engage with malicious content. On the other hand, ham messages often personal or transactional tend to lack these elements, making their absence an implicit indicator of authenticity. Thus, models that can recognize these distinctions are well-positioned to flag suspicious patterns even before analyzing the message content.

However, the rarity of email addresses across all categories observed in fewer than 0.3% of all messages suggests that this feature offers limited discriminatory power. Additionally, structural signals, while powerful, are not foolproof. Sophisticated attackers may deliberately omit or disguise obvious indicators such as URLs or phone numbers to evade detection. As a result, relying solely on structural variables risks underestimating advanced threats. A robust classification strategy must therefore pair these structural patterns with semantic insights drawn from the textual content to ensure a comprehensive and adaptive defense against evolving SMS-based cyberattacks.

Semantic Feature Engineering: TF-IDF

To complement structural insights, we incorporated semantic analysis using TF-IDF (Term Frequency–Inverse Document Frequency). This technique transforms the raw text in the TEXT column into a high-dimensional feature space that captures the relative importance of words within each message class.

Top TF-IDF Terms by Class:

Smishing	Spam	Ham
verify	win	lol
account	150p	ok
urgent	ringtone	gonna

claim	offer	thanks
login	subscribe	home

Interpretation:

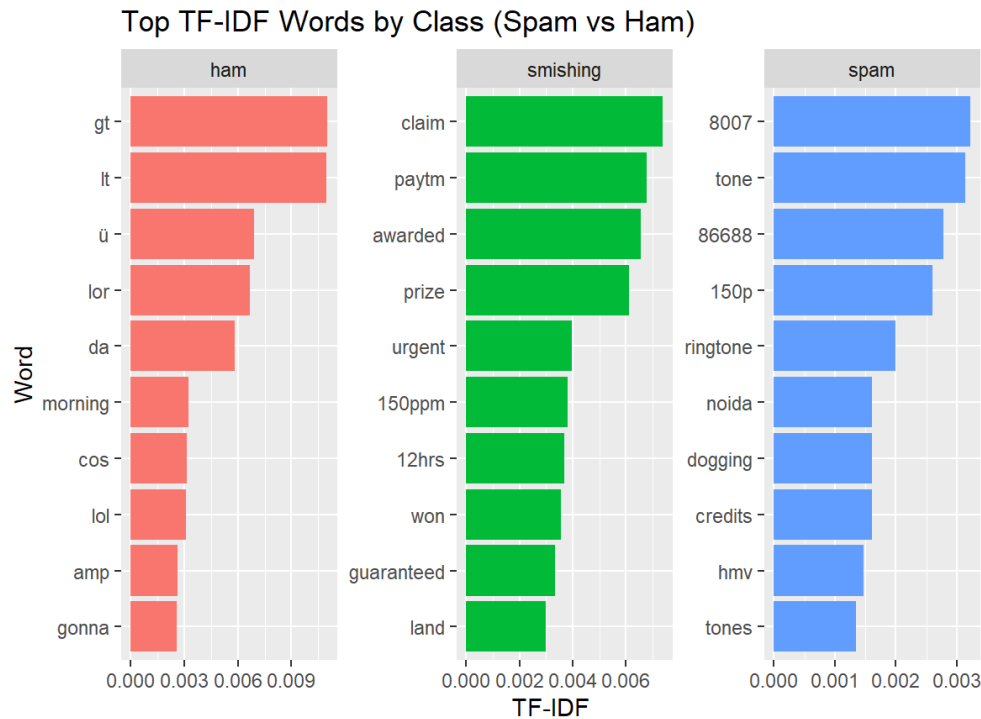
In smishing messages, TF-IDF revealed a high concentration of words associated with urgency, identity verification, and account access terms like “verify,” “account,” and “urgent.” These lexical patterns reflect a common phishing tactic: invoking authority and pressure to elicit an immediate response from the recipient, often leading to data disclosure or unauthorized transactions.

For spam messages, the most prominent TF-IDF terms were promotional and reward-centric, including words like “win,” “150p,” and “ringtone.” This suggests a strategy focused on baiting recipients with contests, subscriptions, or free offers tactics designed to maximize engagement through curiosity or incentive-driven behavior.

In contrast, ham messages were dominated by casual and conversational language. Words such as “lol,” “ok,” and “gonna” reflect natural speech patterns in friendly or familiar exchanges. This informality sets legitimate communication apart from the structured manipulation seen in fraudulent messages, providing a valuable semantic boundary for classification models.

TF-IDF Visualization

Horizontal bar plots of the top TF-IDF terms for each class revealed a clear lexical boundary between smishing/spam and ham. These distinctions allow machine learning models to build internal language profiles for each class and improve classification accuracy, especially in ambiguous or structure-free cases.



Integrated Feature Strategy

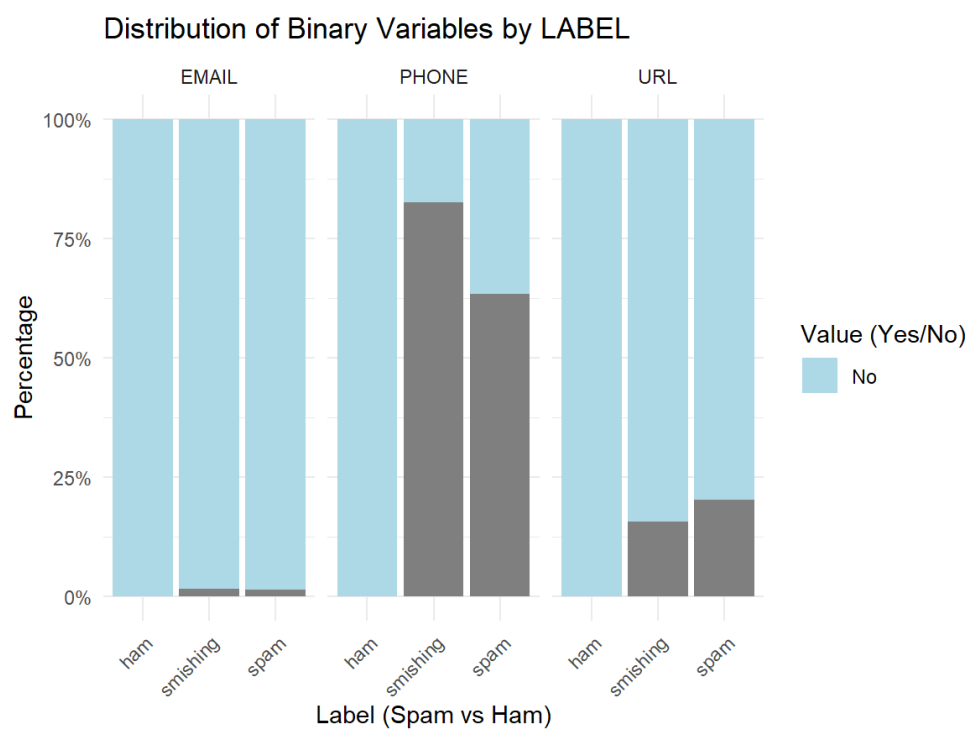
To effectively detect a diverse array of SMS-based cyber threats, we implemented a hybrid feature engineering strategy that merges both structural and semantic insights. Structural features represented by binary indicators for the presence of EMAIL, PHONE, and URL serve as surface-level cues that are frequently exploited by cybercriminals. While simple in form, these metadata variables provide early warning signs of suspicious intent, especially when multiple are used together. For instance, the presence of a URL alongside a phone number significantly increases the likelihood of the message being fraudulent.

Despite the value of these structural cues, many modern phishing attacks are designed to bypass basic pattern recognition. Sophisticated adversaries often avoid overt triggers like URLs or phone numbers, instead relying on psychologically manipulative language that mimics authentic communication. To uncover these subtler threats, we employed TF-IDF vectorization to transform the textual content into a semantic feature space. This allowed

the model to detect messages that, while structurally benign, employed high-risk terms like “verify,” “account,” or “claim” to coax action from recipients.

By integrating both structural and semantic representations, each message was converted into a multidimensional vector profile, enabling deeper contextual understanding. This approach allowed the model not only to catch the blatant fraud attempts, but also to identify more covert and nuanced messages that might otherwise go unnoticed. For example, even if a smishing message lacked a URL, its language profile could still closely match known phishing patterns, prompting accurate classification.

Ultimately, this dual-feature methodology enabled our models to excel in detecting both high-signal and low-signal fraud cases. It recognizes that deception operates on multiple layers visual, structural, and linguistic and that effective detection must reflect this complexity. By synthesizing explicit markers with inferred meaning, we significantly enhanced the robustness and adaptability of our fraud detection framework in the face of evolving attacker behavior.



5. Modeling: Support Vector Machine and Neural Network

To address the challenge of SMS fraud detection, we implemented a supervised learning approach to classify messages into three categories: ham (legitimate), smishing (phishing), and spam (unsolicited promotions). This classification task was approached using two robust machine learning models Support Vector Machine (SVM) and a Neural Network. The choice of these models reflects a balance between interpretability and flexibility: while SVMs are known for their effectiveness in high-dimensional spaces, neural networks offer greater adaptability to complex, nonlinear relationships in the data.

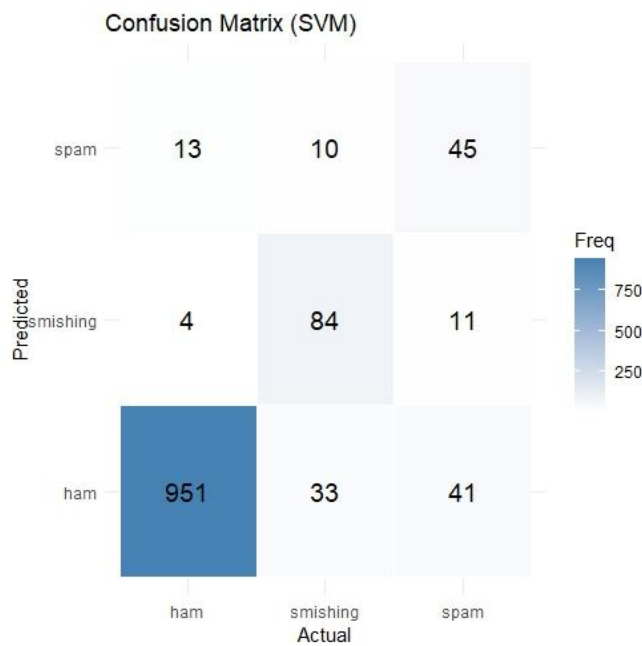
Both models were trained on a comprehensive feature set that combined TF-IDF vectorized representations of the SMS text with binary metadata indicators, specifically the presence of EMAIL, PHONE, and URL within each message. This dual-input architecture allowed the models to leverage both the semantic structure of the language, and the explicit structural cues often associated with fraudulent content. By integrating these feature types, we aimed to enhance model sensitivity to both overtly malicious and subtly deceptive messages.

Support Vector Machine (SVM)

The SVM model was trained using an 80/20 train-test split with a linear kernel, leveraging TF-IDF-transformed text features as input and the LABEL variable as the target. This setup enabled the model to establish decision boundaries in a high-dimensional space, effectively separating message classes based on word importance.

While the SVM demonstrated excellent performance in detecting ham messages, it was less effective in identifying smishing and spam. This imbalance in performance suggests that the linear model may have limitations in capturing the nuanced, nonlinear patterns often present in deceptive or adversarial language.

Confusion Matrix: SVM



The SVM confusion matrix shows strong performance in identifying ham messages, with 951 correct predictions. However, it struggles with smishing detection, correctly identifying only 84 smishing messages and misclassifying 33 as ham. This indicates that while SVM is reliable for non-scam texts, it may miss a significant number of fraudulent messages.

Overall Performance Metrics:

- Accuracy: 90.6%
- 95% Confidence Interval: (88.8%, 92.2%)
- Kappa Statistic: 0.674 (good agreement beyond chance)
- No Information Rate: 81.21%
- McNemar's Test p-value: 3.98×10^{-8} (statistically significant improvement over NIR)

The SVM model achieved an overall accuracy of 90.6%, with a 95% confidence interval ranging from 88.8% to 92.2%, indicating reliable and consistent performance across test

samples. The Kappa statistic of 0.674 reflects good agreement beyond chance, suggesting that the model is making meaningful distinctions between the classes rather than defaulting to the majority label. Additionally, the No Information Rate (NIR) the accuracy that would be achieved by always predicting the most common class was 81.21%, which the SVM significantly outperformed. The low McNemar’s Test p-value (3.98×10^{-8}) confirms that this improvement is statistically significant, affirming the model’s predictive validity.

Class-wise Metrics:

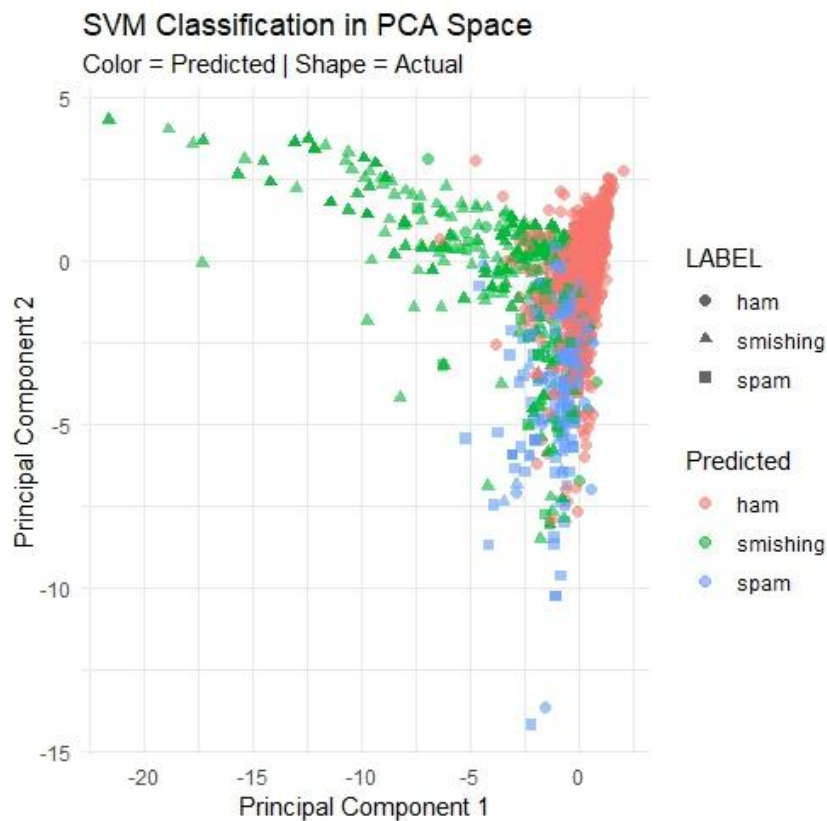
Metric	Ham	Smishing	Spam
Precision	0.928	0.848	0.662
Recall (Sensitivity)	0.982	0.661	0.464
F1 Score	0.954	0.743	0.545
Balanced Accuracy	0.826	0.824	0.721

Insights:

The Support Vector Machine demonstrated outstanding performance in classifying ham messages, with a precision of 92.8% and a recall of 98.2%, resulting in an impressive F1 score of 0.954. This suggests that the model is highly effective at identifying legitimate communication and rarely misclassifies normal user behavior. Such performance is valuable in maintaining user trust and minimizing false positives that could lead to unnecessary message quarantining or alert fatigue.

However, the model’s performance noticeably declined when dealing with smishing and spam, the two most critical fraud classes. Smishing recall was 66.1% and spam recall dropped to 46.4%, indicating that a significant portion of these deceptive messages were incorrectly labeled as benign. While the precision for these categories remained moderate, the relatively low recall poses a risk in real-world applications where false negatives failing to detect an actual phishing attempt can have serious consequences.

This disparity suggests that the SVM's linear decision boundaries may not be flexible enough to capture the more subtle or nonlinear linguistic patterns found in fraudulent messages. Smishing often mimics ham with only slight variations in tone or intent, making it harder to detect using traditional linearly separable models. These limitations point to the potential value of more expressive models, such as neural networks, which can better navigate the nuances of deceptive language and deliver stronger fraud sensitivity.



Neural Network

The neural network model was constructed with a single hidden layer containing 5 neurons and a decay rate of 0.0005 to control overfitting and encourage generalization. It was trained on the full dataset using 5-fold cross-validation, ensuring a reliable assessment of model performance across varying subsets of the data. The feature set mirrored that of the SVM, incorporating 120 TF-IDF-transformed textual features and

binary structural indicators such as EMAIL, PHONE, and URL, allowing the model to learn from both linguistic and structural signals.

What sets the neural network apart is its ability to capture nonlinear dependencies in the data. While linear classifiers like SVM establish boundaries using straight-line separability, neural networks can identify subtle, layered relationships between features. This makes them particularly effective when fraudulent messages are lexically ambiguous or mimic legitimate communication with deceptive intent.

This architectural flexibility allowed the neural network to generalize better across noisy and complex SMS content, especially for smishing detection. In many phishing messages, the difference between a threat and a benign text lies not in obvious structure, but in subtle language cues. The model's ability to detect these cues made it a powerful tool for identifying high-risk messages that evade traditional keyword-based or structure-dependent filters.

Confusion Matrix: Neural Network

Predicted \ Actual	Ham	Smishing	Spam
Ham	940	1	20
Smishing	8	113	33
Spam	20	13	44

Overall Performance Metrics:

- Accuracy: 92.03%
- 95% Confidence Interval: (90.34%, 93.50%)
- Kappa Statistic: 0.756 (strong interclass agreement)
- McNemar's Test p-value: 0.00272 (significant performance over NIR)

The neural network model achieved a notable overall accuracy of 92.03%, with a 95% confidence interval ranging from 90.34% to 93.50%, affirming the model’s consistency and reliability in real-world scenarios. A Kappa score of 0.756 reflects strong interclass agreement, indicating that the model’s predictions go well beyond random chance and effectively differentiate between the three message categories. Importantly, the McNemar’s Test p-value of 0.00272 confirms that the model significantly outperforms a naïve classifier based on the majority class, underscoring its statistical and operational superiority. The confusion matrix further highlights its strength in accurately identifying smishing and ham messages, a critical capability for fraud prevention systems.

Class-wise Metrics:

Metric	Ham	Smishing	Spam
Precision	0.978	0.734	0.571
Recall (Sensitivity)	0.971	0.890	0.454
F1 Score	0.975	0.804	0.506
Balanced Accuracy	0.939	0.926	0.712

Insights:

The neural network delivered exceptional performance in classifying ham messages, achieving a nearly perfect F1 score of 0.975 with precision and recall both exceeding 97%. This strong consistency in identifying legitimate content minimizes false positives and improves overall user experience an essential quality for systems embedded in production environments like telecom services or banking platforms.

Where the neural network truly distinguished itself was in smishing detection, achieving a recall of 89.0% and an F1 score of 0.804. These results are significant given the high cost associated with undetected phishing attempts. The model demonstrated an ability to identify smishing even in the absence of overt structural features like URLs or phone

numbers, highlighting the power of deep pattern recognition through semantic understanding.

Spam detection remained a challenge for the neural network, as seen by its lower recall of 45.4% and a modest F1 score of 0.506. Nonetheless, its overall performance across all classes, particularly its reliability in high-risk fraud categories, it as a more balanced and resilient model than the SVM. Its adaptability to linguistic subtleties makes it a compelling candidate for deployment in real-world fraud detection systems where minimizing false negatives is paramount.

6. Evaluation and Comparison of Models

To evaluate the effectiveness of our classification models, we conducted a detailed analysis of both cross-validation performance and test set outcomes using key metrics such as accuracy, Kappa score, and class-specific F1 scores. Cross-validation ensured that model performance was consistent across different data splits, providing a robust assessment of generalizability. Meanwhile, the test set evaluation offered insight into real-world performance on unseen data, particularly for critical classes like smishing and spam.

This dual-layered evaluation enabled a comprehensive comparison between the Support Vector Machine (SVM) and Neural Network models. By examining not only overall accuracy but also per-class performance, we were able to identify where each model excels and where it struggles. The approach ensured that the chosen model isn't just statistically sound, but operationally effective in detecting high-risk messages with minimal false negatives an essential requirement in SMS fraud detection systems.

Cross-Validation and Test Performance

During 5-fold cross-validation, the neural network achieved an impressive average accuracy of 92.78% and a Kappa score of 0.7769, reflecting strong agreement between predicted and actual labels across multiple training folds. This consistency across splits indicates that the model generalizes well and is not overly dependent on any single subset

of data. When tested on a held-out 20% test set, it sustained this high level of performance with an accuracy of 92.03% and a 95% confidence interval ranging from 90.34% to 93.50%. These metrics provide statistical assurance of the model's reliability in practical deployment scenarios.

The Kappa statistic of 0.756 on the test set further reinforces the model’s predictive value, confirming that it is making meaningful distinctions between classes beyond what would be expected by chance. This is especially important in multi-class classification tasks where imbalanced label distributions such as the prevalence of ham messages can inflate accuracy scores. The neural network’s high Kappa score indicates it maintains balanced performance across all classes, not just the majority.

Despite its strengths, the model exhibited some difficulty in correctly classifying spam messages. Many spam texts were misclassified as either ham or smishing due to linguistic and structural overlap for example, spam often mimics the urgency of smishing or the tone of friendly conversation found in ham. Nevertheless, the neural network delivered exceptional performance on ham (F1 = 0.975) and strong smishing detection (F1 = 0.804), highlighting its capability to identify the most critical fraud patterns while minimizing false negatives in high-risk categories.

Comparative Performance Table

Metric	SVM	Neural Network
Accuracy	90.6%	92.0%
Kappa	0.674	0.756
Smishing F1	0.743	0.804
Spam F1	0.545	0.506
Ham F1	0.954	0.975

Key Insights

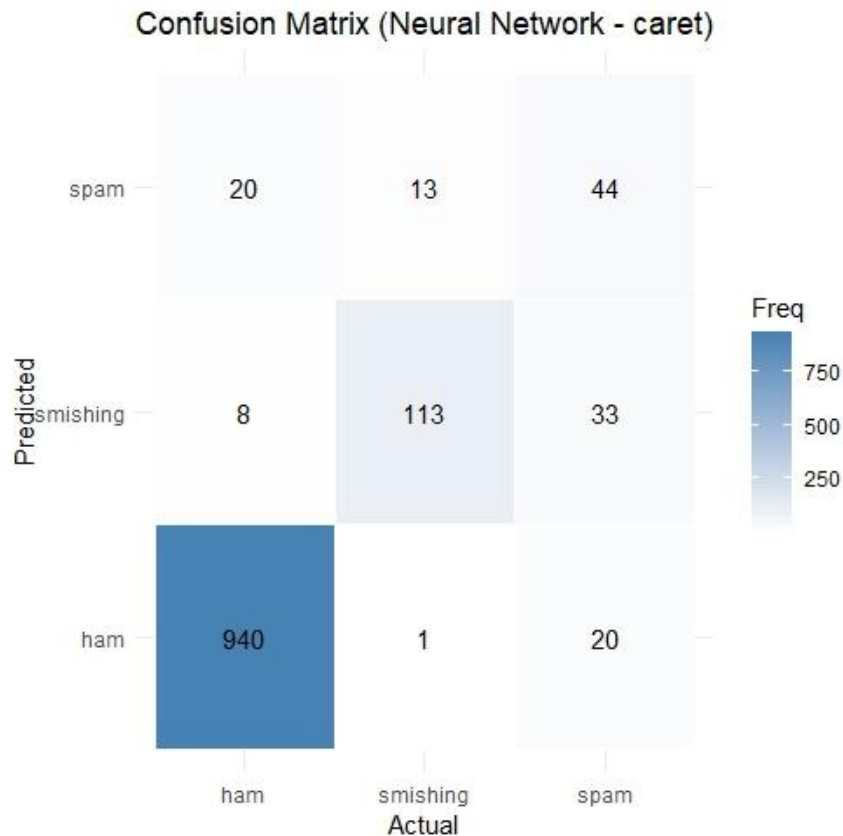
The Support Vector Machine (SVM) stands out for its speed and interpretability, making it suitable for environments with limited computational resources or where decision transparency is essential. It performed exceptionally well in identifying ham messages, achieving high precision and recall. However, its effectiveness declined when detecting smishing and spam, both of which involve subtle and ambiguous language patterns that linear models like SVM struggle to distinguish.

In contrast, the neural network exhibited greater adaptability in handling the complexities of SMS fraud, particularly smishing. Its ability to model nonlinear relationships and capture nuanced semantic signals resulted in higher recall and F1 scores, significantly outperforming SVM in smishing detection. This capability is critical in cybersecurity, where missing a smishing message can mean a direct route to a user's sensitive information. In fact, smishing attacks alone accounted for over \$12.5 billion in reported fraud losses in the U.S. in 2024, highlighting the real-world impact of misclassification.

Both models, however, showed weakness in spam detection, which remains difficult due to its linguistic overlap with both ham and smishing content. This challenge suggests the need for improved feature engineering, class balancing, or ensemble methods to strengthen model performance. Despite this, the neural network consistently proved more reliable and fraud-sensitive. So even though SVM has practical advantages in speed and simplicity, the neural network is ultimately better aligned with the needs of real-world fraud prevention.

Visual Confusion Matrix Analysis

To further validate the neural network's performance, we visualized its classification results using a confusion matrix, which provides a clear snapshot of how well the model differentiates between the three message classes. This tool is especially useful for identifying patterns in misclassification that may not be evident from aggregate metrics alone.



As shown in the matrix, the model excelled at identifying ham messages, correctly classifying 940 out of 961 instances. Its performance on smishing was similarly impressive, with 113 out of 127 messages detected accurately reinforcing the model's ability to spot phishing-related threats with minimal error.

However, the model struggled to distinguish spam messages, correctly identifying fewer than half. This weakness suggests a semantic and structural overlap between spam and other categories, particularly smishing. The results highlight an opportunity for further tuning such as improved feature representation or advanced handling of class imbalance to reduce ambiguity and increase spam detection reliability.

7. Business Implications and Applications

The increasing volume and sophistication of SMS-based scams have elevated the need for real-time fraud detection systems across industries. Our neural network model capable of accurately identifying both overt and covert threats can be deployed within SMS and email gateways to intercept malicious messages before they reach end users. This proactive filtering approach significantly reduces the likelihood of users engaging with phishing content, providing a vital first layer of defense against cyberattacks.

In sectors such as telecommunications and banking, the integration of our model into existing CRM platforms, fraud analytics engines, or mobile security applications can enhance enterprise risk management frameworks. For example, telecom providers can monitor outbound SMS campaigns for regulatory compliance and detect compromised user accounts, while banks can use classification outputs to flag high-risk interactions in customer communication logs.

Beyond detection, the model supports fraud mitigation at scale, especially when paired with alert systems or automated intervention workflows. By flagging suspicious messages in real time, businesses can reduce the incidence of credential theft, unauthorized transactions, and social engineering attacks. Over time, this leads to greater consumer trust, fewer security incidents, and reduced operational costs associated with fraud response and remediation.

From an economic perspective, the stakes are high. According to recent FTC data, the median loss per SMS-based scam is \$1,000, and the total losses in 2024 exceeded \$12.5 billion in the U.S. alone. Scalable AI solutions like the one developed in this project have the potential to prevent millions in annual losses, making them a compelling investment for organizations looking to enhance their cybersecurity posture while protecting customers from the rapidly evolving threat of mobile fraud.

8. Optimization and Future Work

To further improve model performance and adaptability, advanced feature engineering can be applied in future iterations. Incorporating features such as message length, digit counts, use of all-caps, and special characters could offer additional signals indicative of spam or phishing attempts. Furthermore, extracting context-specific keyword patterns, such as commonly used scam phrases or urgency-inducing language, could enhance the semantic richness of the input and increase detection accuracy for more obfuscated fraud attempts.

On the modeling side, the current neural network architecture can be expanded to explore deeper and more expressive models. Using frameworks like Keras or PyTorch, future versions could integrate multiple hidden layers, ReLU activation functions, and dropout regularization to improve learning capacity and reduce overfitting. Additional techniques like batch normalization could stabilize and accelerate training, making the model more scalable and responsive to larger or more diverse datasets.

In terms of evaluation, there is room to enhance the robustness and interpretability of model performance. Future experiments should incorporate ROC-AUC analysis for each class to better visualize true positive rates versus false positives. Moreover, implementing stratified k-fold cross-validation would ensure class balance in every fold, resulting in a more reliable estimate of generalization performance. These techniques will help validate the model's effectiveness, particularly in dealing with class imbalance and edge cases.

Lastly, future development should also consider the computational feasibility and deployment readiness of the models. By tracking training time, memory usage, and inference latency, we can better assess the model's suitability for real-time or mobile environments. Optimization for edge deployment, perhaps through model pruning or quantization, could bring fraud detection directly to the user device, enhancing both privacy and response speed. These steps will help translate academic performance into real-world impact.

9. References

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